KUM DONALSIEN AKWO

UB Junction, Molyko, Buea, Cameroon
donalsienkum@gmail.com
+237 671172400
linkedin.com/in/kumdonalsien

PROFESSIONAL PROFILE

B.Tech Computer Engineering graduate specialising in Networks and Security at the College of Technology, University of Buea, with a strong passion for cybersecurity and cyberspace as a whole. Skilled in cybersecurity, network security engineering, AI/ML-based and hybrid intrusion detection systems, and other network monitoring and auditing tools. Holds 2 years of hands-on technical experience in network installation, configuration, troubleshooting, and security with Telnet Cameroon. Designed and built Bastion IDS, a full hybrid intrusion detection system combining signature-based detection, machine learning ensemble models, deep learning, and unsupervised learning models for real-time threat detection. Proven leader as Program Delegate for the Computer Engineering Department, under the Computer Networks and Systems Maintenance option.

EDUCATION

COLLEGE OF TECHNOLOGY, UNIVERSITY OF BUEA Oct 2023 - Aug 2026

Bachelor of Technology (B.Tech) Computer Engineering (**First Class Honours**) **(Coursework Completed)**

3.73/4.0

GCE ADVANCED LEVEL June 2023 Session

General Certificate of Education (GCE) Board, Cameroon

- Biology, Chemistry, Pure Mathematics with Mechanics, Physics, Computer Science

SKILLS

- Cyber security (Kali Linux, Wireshark, Nmap, Snort, Suricata, IDS/IPS, Ethical Hacking, Penetration Testing, Fortigate, Data Security, Security Operations, Cyber Threat Intelligence, Social Engineering, Information Security)
- Networking (Cisco IOS, Cisco Switches & Routers, OSPF, BGP, IS-IS, RIP, Network Design, Wireless Networking, Network Architecture, IP Addressing)
- Cloud & AWS (AWS EC2, S3, VPC, IAM, CloudFront, CloudWatch, Route 53, VPC design, cloud networking, cloud security)
- Programming (Python, JavaScript, C, MySQL)
- AI & Machine Learning (Scikit-learn, Keras/TensorFlow, Random Forest, XGBoost, CatBoost, Isolation Forest, Autoencoders, Deep Neural Networks, pandas, LLM Evaluation and Benchmarking)
- Frameworks & Tools (FastAPI, Flask, React, Tailwind CSS, Electron, Vite, Docker, Git, VirtualBox, MITRE ATT&CK)
- Operating Systems (Windows, Kali Linux)
- Productivity (Microsoft Office Suite)

EXPERIENCE

TELNET CM August 2024 - September 2024

Intern

- Completed a structured 3-month internship focused on network installation, configuration and security operations
- Gained hands-on experience with physical Cisco switches, routers, Fortigate firewalls and enterprise network infrastructure
- Configured routing protocols (OSPF, BGP, IS-IS, RIP), wireless access points and secure remote access solutions
- Deployed IP-based security surveillance systems and Starlink satellite internet infrastructure on client sites
- Recognised and awarded Best Intern for the batch by TELNET CAMEROON management

TELNET CM October 2024 - Present

Support Staff

- Provides ongoing technical support and network engineering assistance across client projects
- Assists in planning and executing corporate network infrastructure deployments for enterprise and institutional clients
- Applies cybersecurity monitoring best practices and contributes to network security posture assessments
- Supports installation and configuration of Cisco equipment, Fortigate firewalls and IP surveillance systems

TELNET CM July 2025 - September 2025

Senior Intern & Intern Instructor

- Returned for a second formal internship in a senior capacity with expanded technical and leadership responsibilities
- Served as lead instructor and mentor for a new batch of junior interns, delivering training in network configuration, troubleshooting and cybersecurity best practices
- Supervised junior intern lab sessions covering Cisco IOS configuration, routing protocols and network security concepts
- Continued developing advanced skills in corporate network deployments and field security implementations

PROJECTS

Bastion IDS, A Hybrid Intrusion Detection System (2025-2026)

Designed and developed a multi-layer hybrid IDS as a capstone project.

Bastion IDS features a four-layer detection mechanism: Layer 1 is the signature engine (4ET-Open ruleset, 47,357 signatures), Layer 2 is a machine learning ensemble (Random Forest, XGBoost, CatBoost), Layer 3 is a deep neural network specialist (residual MLP), and Layer 4 is an anomaly detection engine. Developed as a standalone desktop application for real-time network monitoring.

Technologies: *Python, FastAPI, React, Scikit-learn, XGBoost, CatBoost, Keras/TensorFlow, Wireshark, Snort*

Stop Arnaque 237, Open Cameroon Scam Dataset (2026)

Created and released an open, labelled dataset of real scams in Cameroon, crowdsourced from the people who receive them and published with a Zenodo DOI under a CC BY 4.0 licence.

Real scam messages sit beside the genuine mobile-money messages they imitate, in English, French and Pidgin, all anonymized and labelled so detection models can tell them apart. Built the full pipeline: ingest, OCR, HMAC pseudonymization, automated leak scanning, labelling and a train/test release in CSV, JSONL and Parquet.

Technologies: *Python, pandas, NLP, Data Anonymization, GitHub Actions, Zenodo, Hugging Face*

Does the Answer Survive Disagreement, LLM Sycophancy Benchmark (2026)

Designed and published an open benchmark that measures whether a language model keeps a correct answer when a user pushes back or asserts a wrong one. The 112-question, four-arm design includes a control arm that separates caving from genuinely updating on evidence. It is graded deterministically, with no LLM judge.

Ran it across 12 models from six labs (Anthropic, Google, OpenAI, xAI, Z.ai and Qwen) on Kaggle Benchmarks, about 16,000 scored calls, parallelized to finish within Kaggle's limits. The write-up was published on DEV for the Kaggle Benchmarking Challenge.

Technologies: *Python, pandas, Concurrency (ThreadPoolExecutor), Data Visualization, Kaggle Benchmarks*

FairGlass, Proof of Fair AI Hiring (2026)

Built on the Midnight blockchain during an MLH hackathon. A Compact smart contract proves that an AI hiring decision followed an agreed fairness policy, issuing a verifiable receipt when the rules hold and refusing one when a forbidden attribute changes the outcome, all without exposing candidate data. Contributed the backend, scoring logic, and the contract bridge.

Technologies: *Python, Flask, Midnight (Compact smart contracts), Cryptographic Commitments*

Evaluation and Recommendation of Network Monitoring Tools (Aug - Sep 2024)

Evaluated industry-standard IDS/IPS and network monitoring tools to identify and prevent cyber threats. Produced formal analysis and recommendations report to enhance organisational security posture.

Technologies: *Wireshark, Snort, Suricata, Kali Linux, Nmap, Security Onion, Kitsune*

Chat Application and File Transfer System (Dec 2024 - Jan 2025)

Built a real-time multi-client chat application with secure file transfer capabilities using Python socket programming.

Technologies: *Python, Socket Programming*

PUBLICATIONS

Tatsachou Ngningaye D., **Kum Donalsien A.**, Gamom Ngounou Ewo R. C., and Boum Alexandre T. "Bastion IDS: Layer Contribution, Threshold Sensitivity, and Consensus Rule Design in a Deployed Hybrid Intrusion Detection System." Journal of Cyber Security Technology (Taylor & Francis Group), 2026. Submitted.

Kum, D. A. (2026). Stop Arnaque 237: Cameroon Scam Reports Dataset (Version 1.0) [Data set]. Zenodo. https://doi.org/10.5281/zenodo.23287618

LANGUAGE

- English (Fluent), French (Basic)

ACTIVITIES

- 237HackFest Capture the Flag (CTF) Competition, organised by Hack The Box Cameroon, gained hands-on experience solving real-world cybersecurity challenges
- Major League Hacking in partnership with the Midnight network, 48-hour hackathon building data protective, privacy-first applications. Developing dApps and tools with ZK technology that put users back in control of their personal data.
- Program Delegate, Computer Engineering Department, Option: Computer Networks and Systems Maintenance, University of Buea, key liaison between students and university administration

SOFT SKILLS

- Leadership
- Teamwork
- Critical Thinking
- Problem Solving
- Effective Communication
- Adaptability
- Time Management
- Attention to Detail
- Resilience

CERTIFICATIONS

- Certified in Cybersecurity (CC), ISC2 (September 2026, valid to September 2029)
- Fortinet Certified Fundamentals (FCF) in Cybersecurity, Fortinet (July 2026, valid to July 2028)
- Cisco Introduction to Cybersecurity, Cisco Networking Academy (September 2025)

- Cisco Networking Academy Capture the Flag 2026, Cisco (May 2026)
- Cyber Crime, Humanitarian Leadership Academy (July 2024)
- Cyber Security, Humanitarian Leadership Academy (July 2024)

ACHIEVEMENTS & AWARDS

- Academic Excellence Recognition, consistently maintaining an excellent CGPA throughout the degree programme
- Leadership skills award from the College of Technology for demonstrating outstanding leadership capabilities in school
- Best Intern Award, Recognised by TELNET CAMEROON management as best intern for the August-September 2024 batch